

AKASH MANOKARAN

B.TECH AI & Data Science Engineer | CGPA: 8.26 / 10

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PROFESSIONAL SUMMARY

Results-driven B.Tech student in Artificial Intelligence and Data Science with strong expertise in Java,python, SQL, Data Structures and Algorithms, Machine Learning, and full-stack web development. Demonstrated ability to design and deliver end-to-end software solutions, IoT systems, and data-driven applications through academic projects. Proven ability to build scalable, data-driven systems using Git version control, object-oriented programming, and database management. Seeking a Software Engineering or Data Science role at a reputed organization to deliver high-impact technical solutions in a fast-paced environment.technical skills.

EDUCATION

Bachelor of Technology (B.Tech) – Artificial Intelligence and Data Science

June 2028

Sri Krishna College of Engineering and Technology (SKCET), Coimbatore, Tamil Nadu

Relevant Coursework: Data Structures, Machine Learning, Database Management Systems, Computer Networks, Operating Systems, Statistics and Probability, Algorithm Design

Diploma – Information Technology | Score: 92.8%

April 2025 | PSG Polytechnic College, Coimbatore

TECHNICAL SKILLS

HARD SKILLS

Programming languages: python, java, c, sql, javascript, html5, css3

Data structures and algorithms: arrays, linked lists, trees, graphs, dynamic programming, binary search, sorting

AI & machine learning: machine learning, data analysis, data-driven application development, iot systems

Web development: html5, css3, javascript, react basics, rest api, spring boot , responsive design

Database management: mysql, oracle sql, relational database design, sql joins, query optimization, crud operations

Software development: object-oriented programming (oop), sdlc, agile methodology, debugging, unit testing basics

Tools & technologies: git, github, vs code, eclipse ide, postman, linux basics, jira basics

Core cs concepts: operating systems, computer networks, dbms, cryptography basics

SOFT SKILLS

problem solving, analytical thinking, team collaboration, technical communication, agile mindset, time management, continuous learning, adaptability

PROJECTS

Student Integrated Platform for Educational Data Management

Tech: HTML, CSS, JavaScript, SQL, Python

Designed and developed a full-stack web portal to centralize educational data management for 500+ students and faculty, improving data accessibility by 40%.

Engineered relational database schema in MySQL with optimized SQL queries, reducing data retrieval time by 35%.

Implemented responsive frontend using HTML5, CSS3, and JavaScript with seamless backend integration, eliminating manual administrative processes.

Followed Software Development Life Cycle (SDLC) practices including requirement analysis, design, development, testing, and deployment.

IoT-Based Real-Time Health Monitoring System

Tech: Arduino C++, Wokwi Simulator, ThingSpeak, Embedded Systems

Designed and simulated a real-time patient health monitoring system using Arduino Uno within the Wokwi environment, integrating pulse rate, temperature, and SpO2 sensor modules with hardware-software communication protocols.

Wrote optimized embedded C++ code for low-latency vital sign data acquisition and processing, achieving real-time health parameter monitoring with less than 100ms sensor response time.

Transmitted live sensor data to ThingSpeak IoT analytics platform via Wi-Fi, enabling cloud-based visualization, trend analysis, and threshold-based health alerts for continuous patient vitals tracking.

SCERTIFICATIONS

Python with Data Science – NPTEL

AWS Cloud Practitioner Essentials – AWS Training badge

JIRA Fundamentals Badge

Algorithm Design Techniques – IMneo

TryHackMe – pentration testing

ACHIEVEMENTS

Scored 92.8% in Diploma in Information Technology – ranked among top 5% performers in the batch.

Successfully solved 100+ Data Structures and Algorithms problems on LeetCode

Basketball Champion— Represented school team and secured championship title, reflecting strong leadership, team collaboration